

Why Relicell is the Right Battery for You

Part of a Rs.1450 crore IT industry group

Wide range to suit any application

VRLA Batteries - 12V, 7AH to 12V, 245AH
2V, 100AH to 2V, 3000AH

Tubular Batteries - 12V, 40AH to 12V, 200AH
2V, 40AH to 2V, 600AH

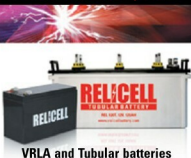
Tested and certified by Central Power Research Institute, Bangalore (CPRI) and Electronic Test and Development Centre (STQC Directorate, Ministry of Information Technology, Govt. of India). Download Test Certificates at www.relicellbattery.com

UL Certified

VRLA Batteries approved and registered in DGS&D for government purchases

54 Service Centres across India - Assured single window service support across India and countrywide installation support.

www.relicellbattery.com



RELICELL

Greenvision Technologies Pvt.Ltd

45/1, 4th Cross, Malleswaram, Bangalore - 560 003.

Tel: +91 080 4245 9459, Fax: +91 080 4245 9460,

Email: info@relicellbattery.com

Award from Ministry

I am really thankful to EFY Tech Center, Delhi for providing me a good experience and knowledge of robotics. I have received the best performance award for robotics in my college from Ministry of Disaster Management and Relief, Rajasthan.

Harshad Shah
Through e-mail

Learning Microcontrollers

I want to learn about AVR and PIC microcontrollers and programs, and develop my own microcontroller-based projects. Is it possible to learn about microcontrollers without microcontroller kits or development boards?

Jawahar K.
Through e-mail

EFY: EFY is a good source to start learning microcontrollers. You can find microcontroller-based projects in its almost every recently-published issue. We have also published a 'Microcontroller-Based Projects' book which includes many AVR- and PIC-based projects.

You need a programmer (PIC or AVR) to burn the code into the chips. You can also purchase an AVR or PIC development board and start learning microcontrollers directly using various examples and applications that you may find on the Internet.

Programmers and development boards can be purchased from Kits'n'Spares. Please visit their website www.kitsnspares.com for more details.

Buying Circuits Online

I want to buy the project 'PC-Based Mobile Robot for Navigation' from 'Circuits on Sale' section of electronicsforu.com website. Will I get a readymade circuit or just the circuit diagram?

Shiladitya Mandal
Through e-mail

EFY: 'Circuit on Sale' section of electronicsforu.com website will provide you only the soft copy of the project in the PDF format. The complete kit (components and PCB) of the mentioned project is not in stock. The availability details

of some of the projects can be found on www.kitsnspares.com website.

Multiple Device Control Using Remote

I want to build a circuit that can switch on and off up to two-three devices, say, light, lamp and fan, individually using a remote. Can anyone help me?

Pankaj Birdi
Facebook

Techila replies: To control these devices using a remote, first you need to decide which remote you are going to use. If it's a standard TV remote, you need to know its protocol to transmit the data. Philips remote uses RC5 protocol. Some of the new ones follow the RC6 protocol. Once you know these protocols, you will be able to receive the data sent by the remote.

You could just buy an RC5 remote for now. Study the RC5 protocol and try to write the code to correctly receive the transmitted signal. I do have the written code (in 'C' language) with me, so I can help you with the code.

Most remotes send data at 36 kHz or 38 kHz. You can use a TSOP1736 or TSOP1738 remote control sensor, respectively, to detect the signal.

Vinit Shandilya replies: Or simply, you can use radio frequency to control the appliances. You can use a Holtek 12E and 12D encoder-decoder pair. It can be easily interfaced to any microcontroller (I have used Atmel's AT89C51).

Arduino Board

Can I make my own Arduino board with discrete components?

Ananth Gopal
Facebook

Vinay Chaddha replies: Yes, it is possible. Check out the 'Open-embedded' page on Facebook for the same. However, making your own board is not cost-effective. Also, an Arduino board is like an Intel processor. You cannot do much with it. You need to add LCD, LED drivers, RTC, IR sensors, etc. Adding on these piggy-backs or wirings will save some money but making a complete project becomes difficult. My advice is to buy a readymade hardware and develop the software.